

Animal Fatal Attraction to Artificial Light

Keren Ighalo and Jack Rosenbaum

Introduction

As human societies expand, so does our reliance on artificial light. Each year since 2011, the sky has become 10% brighter because of anthropogenic light pollution^{1,2,3}. As the presence of artificial light at night (ALAN) surges, the quality of life for animals across taxa worsens. To humans, artificial lighting makes the world clearer, but to non-human species, our lights may be a deadly distraction. Exposure to ALAN can alter circadian rhythms, early-life behaviour, signalling, and mating rituals^{3,4,5}. Approaching artificial light often causes spatial disorientation, temporary blindness, dehydration, collision, or predation, and often results in animal death⁶. For centuries, scientists have documented animals' attraction to light, and insects reliably approach, orbit, and collide with artificial light sources^{7,8,9,10}. It is known that light-attraction is heritable, and there is natural variation in the degree of attraction within species^{11,12,13,14}. The extensive risks associated with approaching artificial light and the heritable nature of light-attraction reflect negative selective pressure against light-attraction. But the continued presence of light-attraction in wild animal populations despite the associated mortality suggests that something is limiting the evolution of light-avoidance.

Behavioural scientists have conferred that ALAN re-entrains the biological rhythms of organisms, disrupting community ecology^{3,15,16}. For instance, fruit flies' (*Drosophila melanogaster*) temperature preference differs with acute exposure to artificial light¹⁷. Other work posits that light-attracted animals, particularly birds, mistake bright white light sources for the moon, which they use as a navigation landmark¹⁸. Further, researchers found correlations between the number of birds killed in collisions with bright structures at night and the presence of the moon^{19,20}. There are several navigational hypotheses; for instance, Fabian et al. (2024) suggested that insects tend to tilt their dorsi towards the strongest light source²². These explanations underestimate birds' remarkable night vision and spatial memory and fail to explain why animals initially approach light sources²¹. In relying on observable behaviours as proxies for innate behaviour, research has overlooked a crucial force in the moth-to-a-flame story: natural selection. Few studies have tried to explain the basis of attraction to ALAN from the perspective of population genomics.

Given the prevalence of light attraction in the natural animal population despite the negative selective pressure on this behaviour in our increasingly urbanized world, I proposed the novel hypothesis that pleiotropy has constrained the evolution of light avoidance. Pleiotropy, whereby one gene affects multiple traits, is common across taxa^{23,24,25}. For instance, a point mutation in a single human gene leads to sickle cell

anemia, which reduces an individual's ability to oxygenate their body, but increases their malaria resistance²⁷. If animals have not evolved light avoidance because of pleiotropic effects that impair fitness, light-avoidance should be associated with reduced fitness. To critically test this prediction, we artificially selected for light-avoidance and tested the fitness of light-avoidant individuals. If a mutation that gives rise to light avoidance also propagates a trait that impedes fitness, then light-attracted individuals should outperform light-avoidant individuals in measurements of fitness. We used fruit flies for this experiment as *Drosophila* show heritable variation in light-attraction and thus can be artificially selected to be light-avoidant or light-neutral¹⁴. When *D. pseudoobscura* were selected for positive and negative phototaxis for 20 generations, there was clear divergence within 15 generations; photopositive lines reached maximum attraction to light, and photonegative lines reached a minimum. Notably, when the experimenters stopped selection, the photonegative lines' behaviour reverted to the baseline faster than that of the photopositive lines²⁶. Taken together, the current literature supports the theory that there is both natural selection against light-attraction in *Drosophila* and a specific unidentified fitness disadvantage associated with light-avoidance. Understanding the root of fatal light-attraction would help us alter our usage and placement of artificial lights, limiting damage to urban animal populations.

Method

Artificial Selection Protocol

We artificially selected for fruit flies that avoid artificial light using 3D-printed light mazes to generate a population of light-avoiding flies that could be tested for negative fitness traits. Each light maze comprised a series of T-shaped choice points where a fly could either move through an opaque funnel (into the dark) or a clear funnel (into the light). After 15 decisions, a fly arrived in 1 of 16 terminal vials, representing their lightscore, or their ratio of light:dark choices. A light-avoidant individual had a lightscore near 1, while a light-attracted individual had a lightscore near 16. A group of up to 150 flies entered each maze at the same time, and the number of flies in each vial was recorded. The top of the light maze had a lid; the paths through opaque funnels were painted dark, and the paths through clear funnels remained bright. A white LED light shone from above during each trial to illuminate the maze's bright paths.

Starting from an ancestral population of 386 founding females, we performed 25 generations of artificial selection. We partitioned a base population into eight lines and assigned flies to selection or control. Within each generation, we tested all selection individuals on lightscore in four sex-separated groups of 150, and the 24 most light-avoiding individuals from each group were selected as breeders for the next generation within each selection line. In each generation, we selected a set of random control individuals as breeders for the next generation within each control line. Every

fifth generation, we tested all control individuals on lightscore in 4 sex-separated groups of 150, and selected 24 random individuals from each group as breeders for the next generation within each control line. Each group was tested in one of four mazes (Maze: A, B, C, D). We maintained selection for 25 generations, and replicate lines were treated as independent evolutionary units. We created four selection lines of light-avoidant fruit flies and four control lines.

Behavioural Assays

After 25 generations of artificial selection, we quantified behavioural responses of all lines on four distinct assays to assess whether reduced light-avoidance is linked to reduced fitness or performance. We used unique flies from generation 26 for each experiment; no individual was tested in more than one assay. We compared the performance of the control and selection lines in four assays: egg-to-adulthood survival, developmental duration, male mating success, and seven-day longevity using methods validated in animal behaviour. We expected selection lines to have lower egg-to-adulthood survivorship and longer developmental duration, poorer mating success, and poorer longevity than control lines.

Egg to Adult Survival and Developmental Duration Protocol

We placed flies from each line in a cage with a petri dish for one hour to mate and lay eggs. We took 100 eggs from each line's Petri dish and placed 20 eggs in each of five vials per line. We recorded the number of eclosed adult flies in each vial three times per day for 14 days.

Male Mating Success Protocol

This protocol combined male-male competition²⁸ with female choice between males. Since males are less sexually selective than females, we did not test for female competitive mating success²⁹. We also ensured all females were between 41 and 51 hours old, as young females are more selective²⁹. In each of 40 vials per line, we placed one 3-day-old control male, one 2-day-old selection male, and one 2-day-old wild female. We recorded the exact start time and whether mating occurred within one hour. If mating did occur, we recorded the identity of the male who mated and the time of mating.

Longevity Protocol

We placed newly eclosed flies in 20 vials per line. Each vial had a group of four males and four females. We recorded the number of deaths of each sex in each vial once per day for seven days.

Statistical Modelling

We completed all our analyses in R version 4.5.3³⁰. We verified all model fits by visually inspecting plots of model residuals using the “DHARMA” package³¹ and visually inspecting diagnostic plots using the “performance” package³². We assessed the significance of fixed effects using the Anova function from the “car” package³³ and calculated estimated marginal means using the “emmeans” package³⁴.

To compare the response to selection between treatments, between sexes, and over time, we fit a GLMM with a Gamma distribution using the “glmmTMB” package³⁵. We used this instead of a normal GLMM as it allows more flexible modelling of complex data. The response variable, lightscore, represents a weighted average score for each group that completes a maze, computed across individuals within the group. Scores were bounded between 1 and 16, with weights reflecting the number of individuals in each of the 16 terminal vials. This model used lightscore as the response variable, with treatment, sex, generation, and their interactions as fixed factors. We also included time of day as a fixed factor since it had only two levels: morning and afternoon. Lineage was nested within treatment, which was nested within generation and included as a random factor. Maze and its position in the test room were also included as random factors. Due to time constraints, we tested control flies every 5th generation. To account for this and simplify our model, we reshaped generation into a categorical variable with five levels using `fct_collapse()` from the “forcats” package³⁶. We used the reshaped generation factor in this model.

To compare egg to adult survival between selected and control flies, we fit a GLMM with a binomial distribution using the “lme4” package³⁷. This model included survival as the response variable, treatment as a fixed factor, and the interaction between treatment, lineage, and vial as a random factor.

To compare the developmental duration of selected and control flies, we fit a GLMM with a gamma distribution using the “lme4” package³⁷. This model included the time it took for flies to eclose as the response variable and treatment and sex as interacting fixed factors. We included the interaction between treatment, lineage, and vial and the interaction between treatment and lineage as random factors.

To compare 7-day survivorship between individuals from the selection and control treatments, we fit a Cox proportional hazards mixed-effects model using the “survival” and “coxme” packages^{38, 39}. This model included days each individual survived as the response variable and treatment and sex as interacting fixed factors. We included the interaction between treatment, lineage, and vial and the interaction between treatment and lineage as random factors.

To compare the mating success of selected males against control males, we fit a generalized linear model (GLM) with a binomial distribution. This model included the mating success of the selected male (only in cases where one of the two males mated) as the response variable, and the intercept and the lineage of the flies as fixed factors.

The lineage of the flies was included as a fixed factor instead of a random factor (in a GLMM) because including it as a random factor resulted in a singular fit.

Results

Artificial Selection

There was a significant treatment x generation interaction (GLMM: Wald $\chi^2_2 = 57.12$, $p < 0.001$), indicating that the response to selection diverged over time between treatments. In early generations, differences between treatments were unclear ($\Delta = -0.0133$, $p = 0.82$), with overlapping confidence intervals (Figure 1b). Treatment differences increased progressively across generations, and by generation 16, there was clear divergence between treatments, with control lines exhibiting consistently higher lightscores than selection lines ($\Delta = 0.5687$, $p < 0.0001$). By generation 16, there was also a detectable difference between selection males and selection females ($\Delta = -0.18$, $p < 0.0001$) that was not apparent in previous generations ($\Delta = -0.07$, $p = 0.28$). By the end of artificial selection in generation 25, treatments had diverged further ($\Delta = 0.57$, $p < 0.0001$), while selection maintained the difference between sexes.

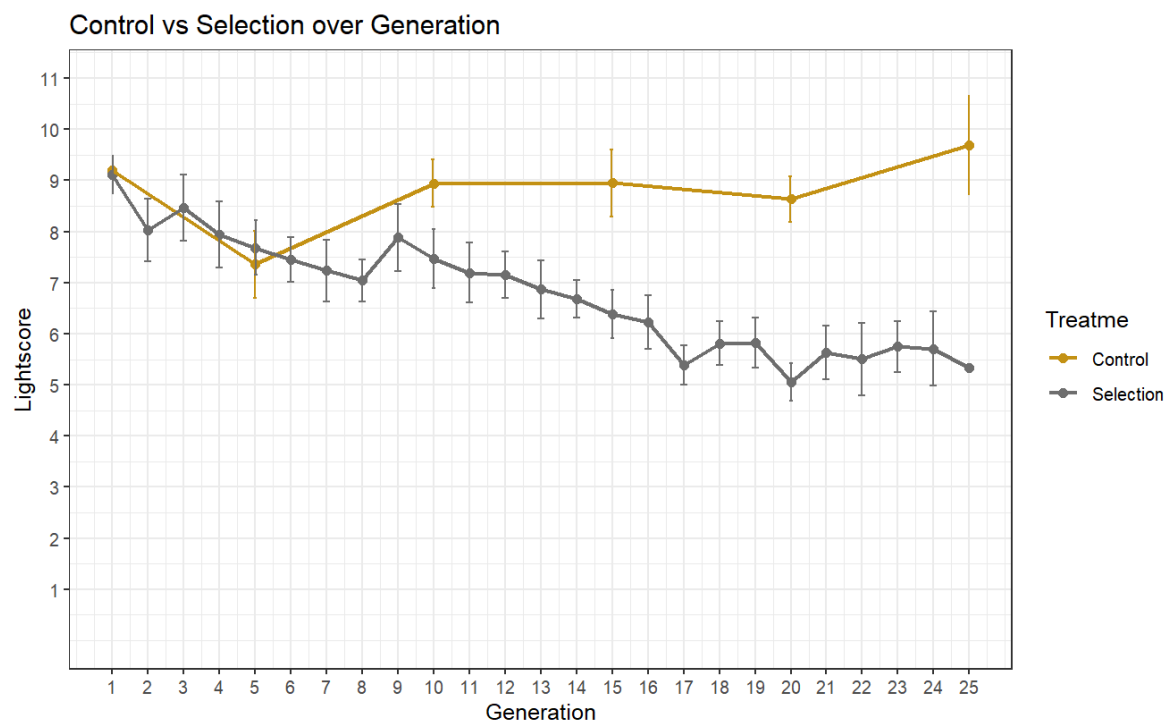


Figure 1a: Lightscore over 25 generations between control and selection treatments, with 95% CI intervals.

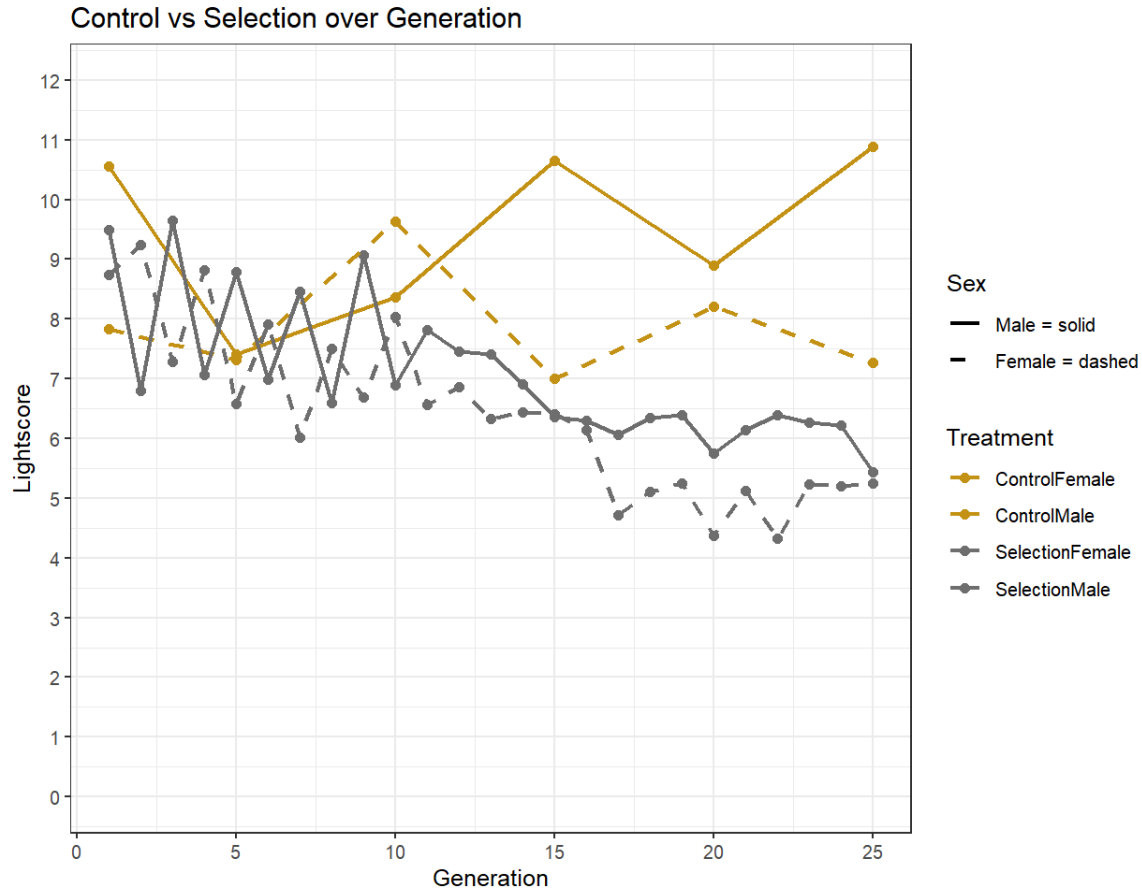


Figure 1b: Lightscore over 25 generations between control and selection males and females.

Egg to Adult Survival & Developmental Duration

We found that significantly more flies from the selection lineages survived to adulthood than flies from the control lineage (GLMM: Wald $\chi^2_2 = 8.135$, $p < 0.01$; Figure 2) but did not detect a difference in development time between selection and control lineages (GLMM: Wald $\chi^2_2 = 0.363$, $p = 0.55$; Figure 3) nor an interaction between sex and treatment (GLMM: Wald $\chi^2_2 = 0.9616$, $p = 0.33$; Figure 3). Consistent with published findings, we found that females developed faster than males (GLMM: Wald $\chi^2_2 = 23.8263$, $p < 0.0001$; Figure 3).

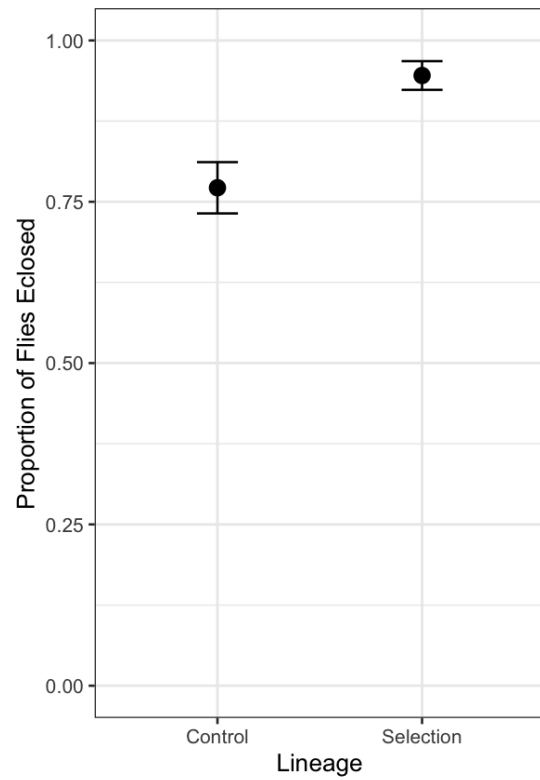


Figure 2: Proportion of flies that eclosed from selected and control lineages.

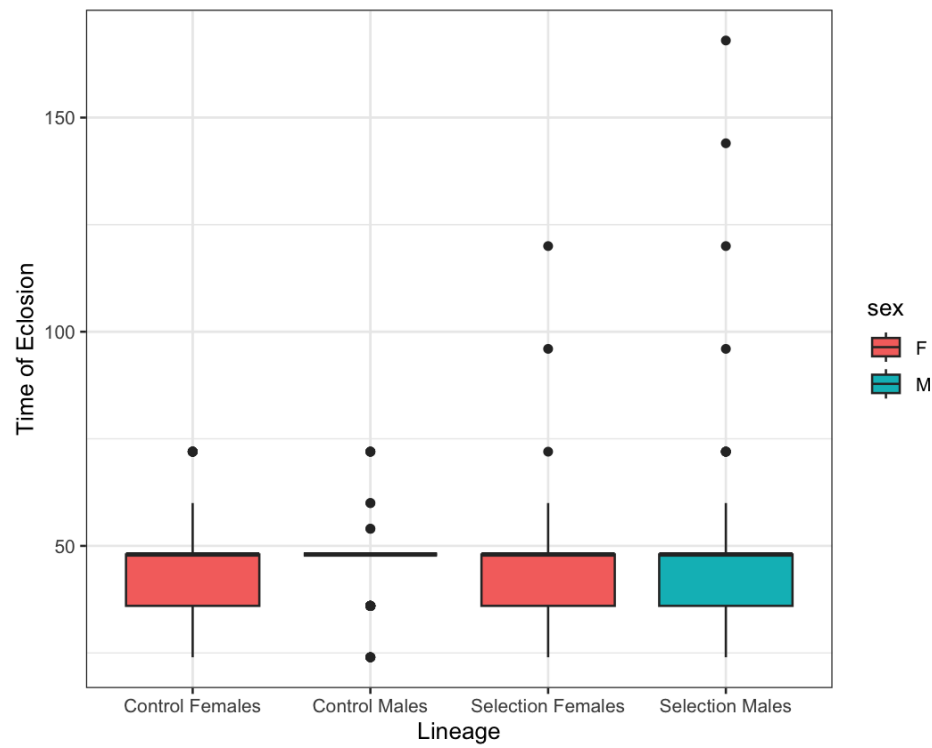


Figure 3: Developmental duration of male and female flies from selection and control lineages.

Male Mating Success

The overall probability of a male from a selection lineage mating over a male from our control lineage was 0.548% (95% CI: 0.478–0.616; Figure 4). We did not detect a meaningful difference in mating success between males from selection and control lineages ($z = 1.36$, $p = 0.17$).

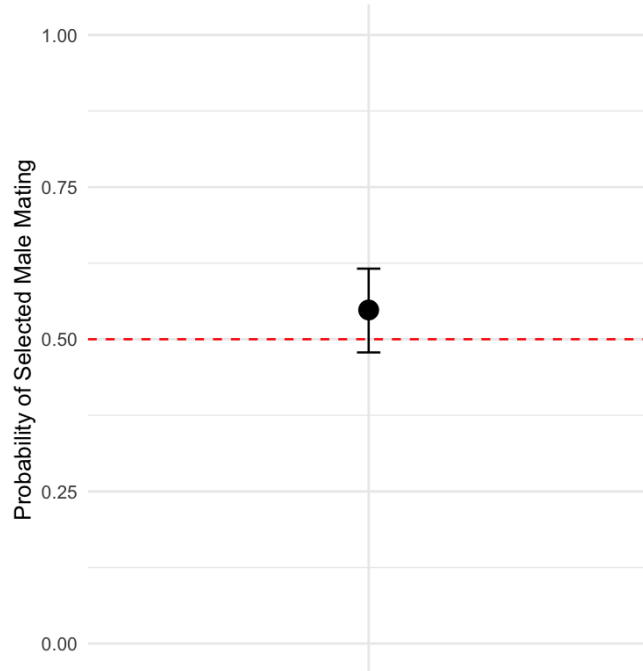


Figure 4: Probability that a male from the selection treatment would secure a mating opportunity over the male from the control treatment.

Longevity

We found that flies from the control lineages died significantly more often than flies from the selection lineage (Cox mixed-effects regression: $\chi^2_2 = 6.05$, $p < 0.05$; Figure 5) and did not detect an effect of sex ($\chi^2_2 = 0.017$, $p = 0.90$) or an interaction between treatment and sex ($\chi^2_2 = 0.939$, $p = 0.33$).

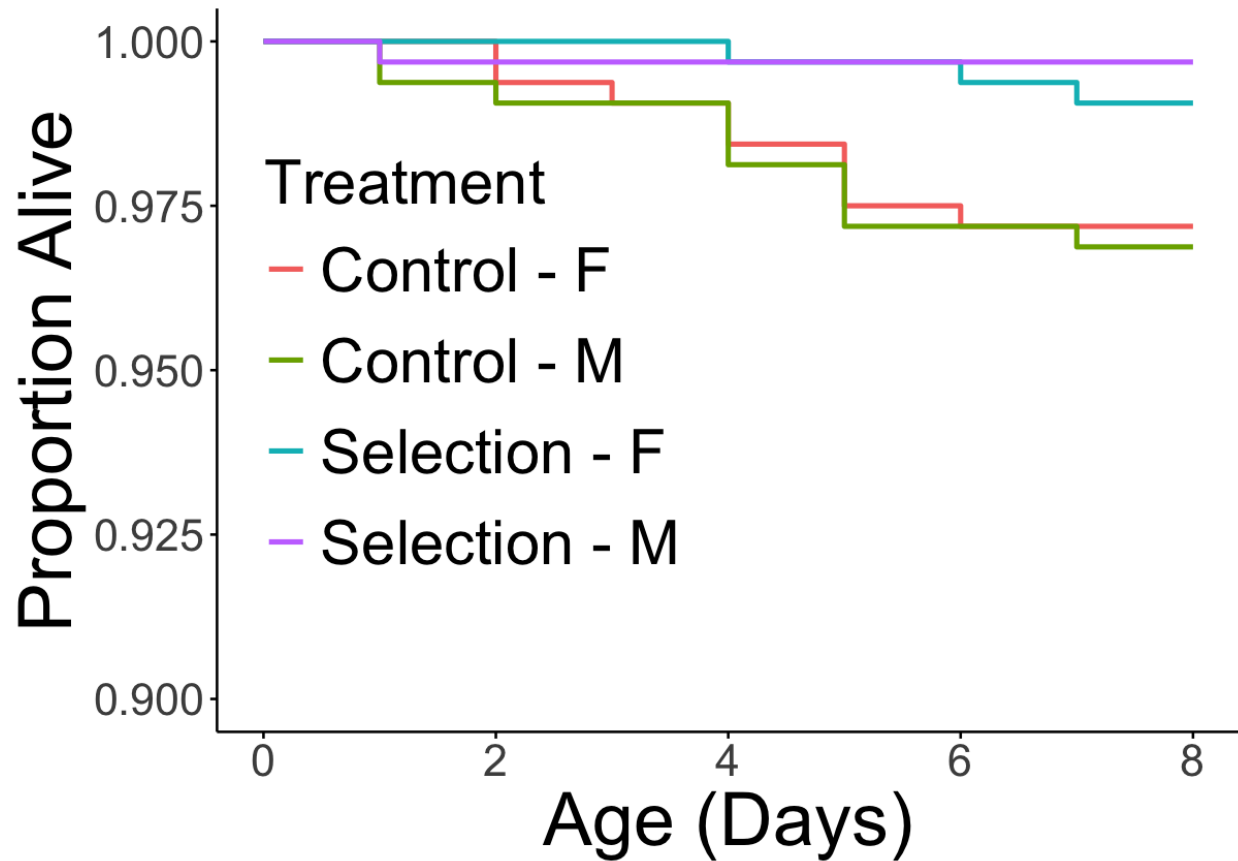


Figure 5: 8-day survival curve showing mortality for flies from the control and selected lineages.

Future Directions

While the population-level analysis of our artificial selection data provides an adequate summary of group differences, it necessitates aggregations across individuals, which makes implicit assumptions about homogeneity within each maze and assumes that relationships between each predictor and the outcome operate at the group level. Aggregation can lead to loss of information and potentially biased inference due to within-group variability. Since we assume each individual's decisions are independent and we are interested in outcomes at the individual level, it makes more philosophical sense to model the data at the individual level. To do this, we converted our data from one maze per row to one vial per row, then one fly per row. We created a binomial model using `glmmTMB`³⁵; `lightscore` is a proportion of light to dark decisions. This model had the same fixed and random effects as our population-level model, but the response variable was the fly's individual vial ID.

In addition to our four behavioural assays, we also measured each line's circadian rhythm in generation 26. If light-avoidance is associated with negative fitness effects, one possibility is that light-avoiding flies have differences in circadian rhythm compared to control (light-neutral or light-attracted) flies. At 9:00 on the morning of each

test day, we aspirated a single fly into a small snap-cap vial (22 mm diameter × 48 mm long) and placed the vials into 1 of 2 *Drosophila* activity monitors (DAM) (Trikinetics Inc.; software version 3.10). We placed each monitor in one of two opaque plastic containers with an LED lightbulb over a hole in the center of each container lid to illuminate the monitors from above. This light turned on at 21:00 and off at 9:00. The DAM began recording as soon as the lid was closed and the flies were in place. We left the flies to acclimatize from 9:00 to 21:00. The DAM automatically recorded activity as the number of times that each fly crossed the ring of infrared sensors that surrounded each snap-cap vial during each minute of the 24-hour test period (21:00 to 21:00). In total, we assayed 16 males and 16 females from each line. Each line pair (S1 C1, S2 C2, etc) completed the assay on a unique day. Analyzing the data from this experiment would allow us to evaluate the link between circadian rhythm and behaviour in the presence of artificial light. To do so, we will compare activity between populations using linear mixed models from lme4³⁷. To account for the effects of circadian rhythm on activity, we propose a parametric model from glmmTMB³⁵ that imposes circadian periodicity to represent time of day over 24 hours. Alternatively, we may fit a natural smooth cubic spline with 5 knots. This more elastic approach would allow for irregular patterns in behaviour with only a few parameters. Since we expect the light-avoidant lines to have abnormal activity patterns, we want the model curve to have a lot of flexibility. Our fixed effects will be treatment, sex, and monitor. Our random effects will be lineage (nested within treatment) and day.

References

1. Kyba, C. C. M., Altıntaş, Y. Ö., Walker, C. E., & Newhouse, M. (2023). Citizen scientists report global rapid reductions in the visibility of stars from 2011 to 2022. *Science*, 379(6629), 265–268.
2. Kyba, C. C. M., Tong, K. P., Bennie, J., Birriel, I., Birriel, J. J., Cool, A., Danielsen, A., Davies, T. W., Outer, P. N. den, Edwards, W., Ehlert, R., Falchi, F., Fischer, J., Giacomelli, A., Giubbilini, F., Haaima, M., Hesse, C., Heygster, G., Hölker, F., ... Gaston, K. J. (2015). Worldwide variations in artificial skyglow. *Scientific Reports*, 5(1), 8409.
3. Swaddle, J. P., Francis, C. D., Barber, J. R., Cooper, C. B., Kyba, C. C. M., Dominoni, D. M., Shannon, G., Aschehoug, E., Goodwin, S. E., Kawahara, A. Y., Luther, D., Spoelstra, K., Voss, M., & Longcore, T. (2015). A framework to assess evolutionary responses to anthropogenic light and sound. *Trends in Ecology & Evolution*, 30(9), 550–560.
4. Brown, T. M., Wilhelm, S. I., Slepkov, A. D., Baker, K., Mastromonaco, G. F., & Burness, G. (2024). Navigating the night: Effects of artificial light on the behaviour of Atlantic puffin fledglings. *Animal Behaviour*, 218, 135–148.
5. Gaston, K. J., Davies, T. W., Nedelec, S. L., & Holt, L. A. (2017). Impacts of artificial light at night on biological timings. *Annual Review of Ecology, Evolution, and Systematics*, 48, 49–68.

6. Owens, A. C. S., & Lewis, S. M. (2018). The impact of artificial light at night on nocturnal insects: A review and synthesis. *Ecology and Evolution*, 8(22), 11337–11358.
7. Lalagüe, H., & Pétilon, J. (2023). Light attraction hypothesis in Arachnids: A new test in neotropical forests. *The Journal of Arachnology*, 51(2), 111–113.
8. Wilhelm, S. I., Schau, J. J., Schau, E., Dooley, S. M., Wiseman, D. L., & Hogan, H. A. (2013). Atlantic Puffins are Attracted to Coastal Communities in Eastern Newfoundland. *Northeastern Naturalist*, 20(4), 624–630.
9. Wilson, R., Cooper, C. E. C., Meah, R. J., Wakefield, A., Roberts, N. W., & Jones, G. (2023). The spectral composition of a white light influences its attractiveness to *Culex pipiens* mosquitoes. *Ecology and Evolution*, 13(1), e9714.
10. Darwin, C. 1871. Darwin Correspondence Project, Letter no. 7692 <https://www.darwinproject.ac.uk/letter/?docId=letters/DCP-LETT-7692.xml>
11. Altermatt, F., & Ebert, D. (2016). Reduced flight-to-light behaviour of moth populations exposed to long-term urban light pollution. *Biology Letters*, 12(4), 20160111.
12. Benzer, S. (1967). Behavioral mutants of *Drosophila* isolated by countercurrent distribution. *Proceedings of the National Academy of Sciences*, 58(3), 1112–1119.
13. Czaczkes, T. J., Bastidas-Urrutia, A. M., Ghislandi, P., & Tuni, C. (2018). Reduced light avoidance in spiders from populations in light-polluted urban environments. *The Science of Nature*, 105(11), 64.
14. Hadler, N. M. (1964). Heritability and phototaxis in *drosophila melanogaster*. *Genetics*, 50(6), 1269–1277.
15. Hölker, F., Bolliger, J., Davies, T. W., Giavi, S., Jechow, A., Kalinkat, G., Longcore, T., Spoelstra, K., Tidau, S., Visser, M. E., & Knop, E. (2021). 11 Pressing Research Questions on How Light Pollution Affects Biodiversity. *Frontiers in Ecology and Evolution*, 9.
16. Sanders, D., Frago, E., Kehoe, R., Patterson, C., & Gaston, K. J. (2021). A meta-analysis of biological impacts of artificial light at night. *Nature Ecology & Evolution*, 5(1), 74–81.
17. Head, L. M., Tang, X., Hayley, S. E., Goda, T., Umezaki, Y., Chang, E. C., Leslie, J. R., Fujiwara, M., Garrity, P. A., & Hamada, F. N. (2015). The Influence of Light on Temperature Preference in *Drosophila*. *Current Biology*, 25(8), 1063–1068.
18. Baker, R. (1984). *Bird Navigation: The Solution of a Mystery?* Cambridge, UK: Cambridge University Press.
19. Verheijen, F.J. (1980). The Moon: a neglected factor in studies of collisions of nocturnal migrant birds with tall lighted structures and with aircraft. *Die Vogelwarte* 30, 305-320.

20. Crawford, R. L. (1981b). Bird Kills at a Lighted Man-made Structure: Often on Nights Close to a Full Moon.
21. Niu, X., Jiang, Z., Peng, Y., Huang, S., Wang, Z., & Shi, L. (2022). Visual cognition of birds and its underlying neural mechanism: A review. *Avian Research*, 13, 100023.
22. Fabian, S. T., Sondhi, Y., Allen, P. E., Theobald, J. C., & Lin, H.-T. (2024). Why flying insects gather at artificial light. *Nature Communications*, 15(1), 689.
23. Casares, P., Carracedo, M., & García-Florez, L. (1997). Analysis of larval behaviours underlying the pupation height phenotype in *Drosophila simulans* and *D melanogaster*. *Genetics Selection Evolution*, 29, 589.
24. Conner, J. K., Cooper, I. A., La Rosa, R. J., Pérez, S. G., & Royer, A. M. (2014). Patterns of phenotypic correlations among morphological traits across plants and animals. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 369(1649), 20130246.
25. Sokolowski, M. B. (2001). *Drosophila*: Genetics meets behaviour. *Nature Reviews Genetics*, 2(11), 879–890.
26. Dobzhansky, T., & Spassky, B. (1969). Artificial and natural selection for two behavioral traits in *Drosophila pseudoobscura*. *Proceedings of the National Academy of Sciences*, 62, 75–80.
27. Hedrick, P. W. (2011). Population genetics of malaria resistance in humans. *Heredity*, 107(4), 283-304. <https://doi.org/10.1038/hdy.2011.16>
28. Baxter, C., Mentlik, J., Shams, I., & Dukas, R. (2018). Mating success in fruit flies: Courtship interference versus female choice. *Animal Behaviour*, 138, 101-108. <https://doi.org/10.1016/j.anbehav.2018.02.010>
29. Dukas, R. (2004). Male fruit flies learn to avoid interspecific courtship. *Behavioral Ecology*, 15(4), 695-698. <https://doi.org/10.1093/beheco/arh068>
30. R Core Team (2025). *_R: A Language and Environment for Statistical Computing_*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org>
31. Hartig F (2026). *DHARMA: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models*. R package version 0.5.0, <https://github.com/florianhartig/dharma>.
32. Lüdtke D, Ben-Shachar M, Patil I, Waggoner P, Makowski D (2021). “performance: An R Package for Assessment, Comparison and Testing of Statistical Models.” *Journal of Open Source Software*, 6(60), 3139. doi:10.21105/joss.03139.
33. Fox J, Weisberg S (2019). *An R Companion to Applied Regression*, Third edition. Sage, Thousand Oaks CA. <https://www.john-fox.ca/Companion/>.

34. Lenth R, Piaskowski J (2026). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package version 2.0.3, <https://rvlenth.github.io/emmeans/>.
35. Mollie E. Brooks, Kasper Kristensen, Koen J. van Benthem, Arni Magnusson, Casper W. Berg, Anders Nielsen, Hans J. Skaug, Martin Maechler and Benjamin M. Bolker (2017). *glmmTMB Balances Speed and Flexibility Among Packages for Zero-inflated Generalized Linear Mixed Modeling*. The R Journal, 9(2), 378-400. doi: 10.32614/RJ-2017-066.
36. Wickham H (2025). *forcats: Tools for Working with Categorical Variables (Factors)*. R package version 1.0.1, <https://forcats.tidyverse.org/>.
37. Bates D, Mächler M, Bolker B, Walker S (2015). "Fitting Linear Mixed-Effects Models Using lme4." *Journal of Statistical Software*, **67**(1), 1–48. doi:10.18637/jss.v067.i01.
38. Therneau T (2026). *A Package for Survival Analysis in R*. R package version 3.8-6, <https://CRAN.R-project.org/package=survival>.
39. Therneau TM (2024). *coxme: Mixed Effects Cox Models*. R package version 2.2-22, <https://CRAN.R-project.org/package=coxme>.